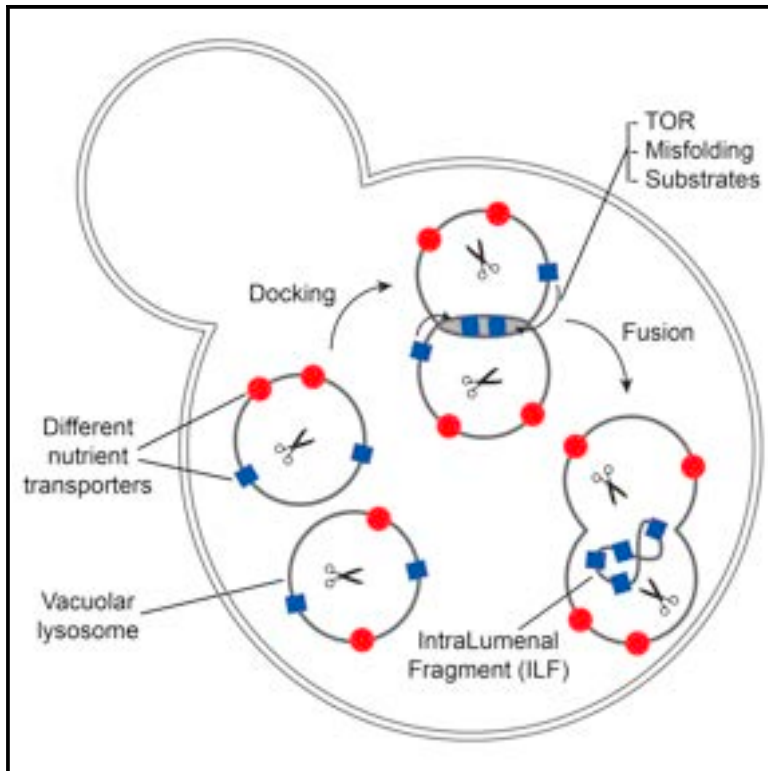


Developmental Cell

Selective Lysosomal Transporter Degradation by Organelle Membrane Fusion

Graphical Abstract



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In Brief

Lysosomal nutrient transporters are crucial to cellular function. McNally et al. discover that they are regulated by selective sorting, during lysosomal fusion, into an area between apposing membranes that is then internalized and degraded. This ESCRT-independent process mediates protein quality control, responds to substrate levels, and is stimulated by TOR signaling.

Highlights

- Lysosomal polytopic proteins are sorted and degraded upon organelle membrane fusion
- This ESCRT-independent process requires the fusion machinery for protein sorting
- This process mediates protein quality control and responds to substrate levels and TOR
- Called the ILF pathway, this process underlies lysosome homeostasis and remodeling



Selective Lysosomal Transporter Degradation by Organelle Membrane Fusion

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SUMMARY

Lysosomes rely on their resident transporter proteins to return products of catabolism to the cell for reuse and for cellular signaling, metal storage, and maintaining the luminal environment. Despite their importance, little is known about the lifetime of these transporters or how they are regulated. Using *Saccharomyces cerevisiae* as a model, we discovered a new pathway intrinsic to homotypic lysosome membrane fusion that is responsible for their degradation. Transporter proteins are selectively sorted by the docking machinery into an area between apposing lysosome membranes, which is internalized and degraded by luminal hydrolases upon organelle fusion. These proteins have diverse lifetimes that are regulated in response to protein misfolding, changing substrate levels, or TOR activation. Analogous to endocytosis for controlling surface protein levels, the “intraluminal fragment pathway” is critical for lysosome membrane remodeling required for organelle function in the context of cellular protein quality control, ion homeostasis, and metabolism.

INTRODUCTION

All eukaryotic organisms use organelles called lysosomes to recycle biomaterials. To receive incoming biomaterial, lysosomes fuse with membrane encapsulated compartments containing proteins, DNA, damaged organelles, or cellular debris packaged within cells through autophagy or collected from outside cells by endocytosis. Fusion events expose the biomaterials to luminal hydrolases for catabolism and once degraded, resident lysosomal transporters translocate the constituents from the lumen to the cytoplasm for reuse by the cell. These include transporters for nucleotides (ENT3; [Song et al., 2014](#)), amino acids (SLC38A9; [Jung et al., 2015](#)), lipids (NPC1; [Berger et al., 2005](#)), or metabolites and macromolecules (e.g., ABCB6; [Kiss et al., 2015](#)). Lysosomes are also critical for storage of divalent metals, including zinc by ZnT2 ([Hennigar and Kelleher, 2015](#)), copper by SLC31A2 ([Schweigel-Rontgen, 2014](#)), and iron by NRAMP2/DMT1 ([Ehrnstorfer et al., 2011](#)), essential co-factors for many cellular enzymes. Because of their importance

to lysosome physiology, mutations that target many of these transporters underlie human lysosomal storage disorders, such as mucopolipidosis type IV (TRPML1; [Waller-Evans and Lloyd-Evans, 2015](#)) or Niemann-Pick disease type C (NPC1; [Berger et al., 2005](#)). Despite being critical for lysosome function and their relationship with human disease, we know little about how these transporters are regulated or degraded.

Most polytopic proteins, after passing ER quality control, are degraded by the multivesicular body (MVB) pathway. This pathway mediates the ubiquitin-dependent downregulation of surface transporters and receptors ([Katzmann et al., 2001](#); [Babst, 2011](#)) either constitutively (e.g., Ste3; [Davis et al., 1993](#)) or in response to cellular cues, such as changes in substrate levels (e.g., Mup1; [MacDonald et al., 2012](#)) or upon protein misfolding (e.g., Fur4; [Keener and Babst, 2013](#)). A critical step in the MVB pathway is the inward budding of the membrane, which results in formation of intraluminal vesicles (ILVs) within the MVB. Importantly, surface proteins targeted for downregulation are sorted into the ILVs where, upon heterotypic fusion between the MVB and the lysosome, the protein-laden ILVs are delivered to the lysosomal lumen for exposure to acid hydrolases and degradation ([Katzmann et al., 2001](#); [Shields and Piper, 2011](#)).

MVB formation and protein sorting are achieved through the sequential function of five endosomal sorting complexes required for transport (ESCRTs; [Babst, 2011](#)). Recently, Emr and colleagues demonstrated that some resident lysosomal membrane proteins utilize ESCRTs for degradation in *Saccharomyces cerevisiae* ([Li et al., 2015a, 2015b](#)). While this is the first description of a mechanism for lysosomal polytopic protein degradation, the authors disclose that this process, termed vRED (vacuolar membrane recycling and degradation), does not regulate the lifetimes of some transporters and pumps, e.g., Vph1 (subunit of the V-ATPase), Fth1 (iron transporter), and Vba4 (cationic amino acid transporter). In addition, cells survive and divide when key components of the ESCRT machinery are eliminated or dysfunctional ([Mageswaran et al., 2015](#)). Thus eukaryotic cells likely have an ESCRT-independent pathway to degrade polytopic proteins. If so, this new degradation pathway must possess mechanisms that sort labeled polytopic proteins into an area of membrane that is internalized within the lysosomal lumen, exposing them to hydrolases for degradation. Based on current knowledge, there is a cellular event that may fulfill all criteria: the process of homotypic lysosome membrane fusion.

Lysosomes are dynamic organelles constantly undergoing cycles of fission and homotypic fusion to retain copy number and accommodate organelle inheritance ([de Duve and Wattiaux, 1966](#); [Wickner and Haas, 2000](#)). Homotypic vacuolar lysosome

fusion requires a sequential series of four distinct subreactions: priming, tethering, docking, and fusion. Priming involves the disassembly of *cis*-SNARE complexes, which ensures that individual SNARE proteins will interact to form new complexes in *trans*, a prerequisite for fusion. Tethering occurs when apposing membranes make first contact, which requires active Ypt7, a Rab-GTPase, and the HOPS (homotypic vacuole fusion and protein sorting) tethering holocomplex. Docking is operationally defined as the period between tethering and lipid bilayer fusion. During this stage, fusogenic lipids and proteins assemble into an expanding ring that stabilizes contact at the vertex between apposing membranes (Wickner and Haas, 2000; Wang et al., 2002; Fratti et al., 2004). Here, *trans*-SNARE complexes within this assembly progressively twist into stable helical bundles that drive lipid bilayers together, leading to complete membrane fusion. Importantly, the area of membrane and embedded proteins within the vertex ring, called the boundary, is internalized as a byproduct of this fusion event (Wang et al., 2002). Topologically, formation of this intraluminal fragment is analogous to genesis of intraluminal vesicles in the MVB pathway, which accommodates exposure of membrane proteins to hydrolases for degradation. Although fusion proteins are spared upon fusion, it was previously unclear whether lysosomal polytopic proteins can be selectively sorted into the boundary. If so, this process could accommodate their internalization into the lumen for subsequent degradation. Thus we hypothesize that resident lysosomal membrane proteins may be selectively sorted into the boundary, internalized, and degraded upon homotypic vacuole fusion.

RESULTS

Polytopic Proteins Are Selectively Sorted for Degradation during Homotypic Lysosome Fusion

To test our hypothesized model of lysosomal polytopic protein degradation (Figure 1A) we examined the sorting (membrane distribution), internalization, and degradation of eight GFP-tagged lysosomal polytopic proteins during homotypic fusion events using *S. cerevisiae* and its vacuolar lysosome as models, as the underlying machinery and transporters are largely evolutionarily conserved. To begin with, we monitored the distribution of GFP-tagged transporter proteins on the vacuole membrane over time during homotypic vacuole fusion within live cells (Figures 1B and 1C; Movies S1, S2, and S3). We first confirmed that Vph1, the stalk domain of the V-ATPase complex, was uniformly distributed on the lysosomal membrane during fusion and, as a consequence, internalized (Wang et al., 2002). Then we made two striking observations: (1) the lysosomal multicopper oxidase Fet5 (Urbanowski and Piper, 1999) was excluded from the boundary membrane of docked organelles and spared from being internalized upon fusion, despite formation of a luminal lipid membrane fragment; (2) the lysosomal iron transporter Fth1 accumulated within the boundary and was internalized upon lysosome fusion. We studied five other GFP-tagged resident polytopic proteins (Figure 1D) and classified their sorting phenotypes as either uniformly distributed (Vph1), excluded from (Fet5; Ybt1 and Ycf1, ABC-C transporters; the sterol transporter Ncr1), or enriched within (Fth1; the zinc transporter Cot1; and Sna4, an E3-Ub ligase adaptor protein) the boundary mem-

brane during lysosome fusion. These data demonstrate that protein sorting is selective and highlights the dependence of this process on intraluminal membrane fragment formation during organelle fusion. As such, we named this new cellular protein degradation pathway the intraluminal fragment (ILF) pathway.

To further study the molecular basis of the ILF pathway, we isolated vacuolar lysosomes from yeast cells and conducted *in vitro* fusion reactions (Wickner, 2010). We first confirmed our initial findings, whereby the same sorting phenotypes were observed *in vitro* as *in vivo* for all GFP-tagged polytopic proteins studied (Figure 2A). This important result indicates that all molecular machinery necessary for protein sorting and internalization during fusion copurifies with the organelles. Next, to examine the sorting process in more detail, we quantified GFP fluorescence within the boundary on organelles undergoing fusion *in vitro*. Compared with Vph1-GFP fluorescence, which is uniformly distributed on the boundary and outer membranes (Figure 2B), Fet5-GFP is expelled from the boundary, whereas Fth1-GFP concentrates within apposed boundary membranes as the fusion reaction progresses over time (Figure 2C). We next calculated GFP fluorescence within the boundary membrane relative to the outside membrane at all organellar contact sites observed (Figure 2D) and found that boundary membranes contained more enriched proteins (e.g., Fth1-GFP) and fewer excluded proteins (Fet5-GFP) compared with Vph1-GFP, confirming the sorting phenotypes for all polytopic proteins studied.

Results from these experiments also offer evidence suggesting that boundary-localized polytopic proteins are internalized (Figures 2E and 2F): compared with Vph1-GFP there is no luminal accumulation of Fet5-GFP (even at 120 min when most vacuoles have fused; Merz and Wickner, 2004) but more Fth1-GFP accumulated in the lumen as the fusion reaction progresses, consistent with their sorting phenotypes. When internalized, polytopic proteins embedded within the ILF are exposed to hydrolases for degradation. Thus, we conducted western blot analysis to detect cleavage of GFP from the polytopic proteins by luminal proteases (Figure 2G; Li et al., 2015a). Consistent with results from our micrographic analysis, less GFP was cleaved from Fet5 and more was cleaved from Fth1 compared with Vph1 after fusion. This approach also allowed us to accurately measure rates of protein turnover, as it eliminates the impact of new protein delivery to the isolated lysosomal vacuoles during study, since polysomes and components of the secretory system are absent from the preparation (Cabrera and Ungermann, 2008). Deleting the *PEP4* and *PRB1* genes encoding luminal proteases prevented GFP cleavage, further validating our model of the ILF pathway and highlighting its ability to selectively control lysosomal protein lifetimes. Furthermore, it suggests that lysosomes, and possibly other organelles, have the capability to remodel the protein landscape within their membranes with every fusion event. Two questions immediately arose upon making this exciting discovery: what is the function of the ILF pathway, and what mechanisms underlie this process?

The ILF Pathway Mediates Quality Control of Lysosomal Polytopic Proteins

A primary function of cellular protein degradation pathways is to clear misfolded or damaged proteins, a vital quality control mechanism within all eukaryotic cells (Chen et al., 2011; Babst,

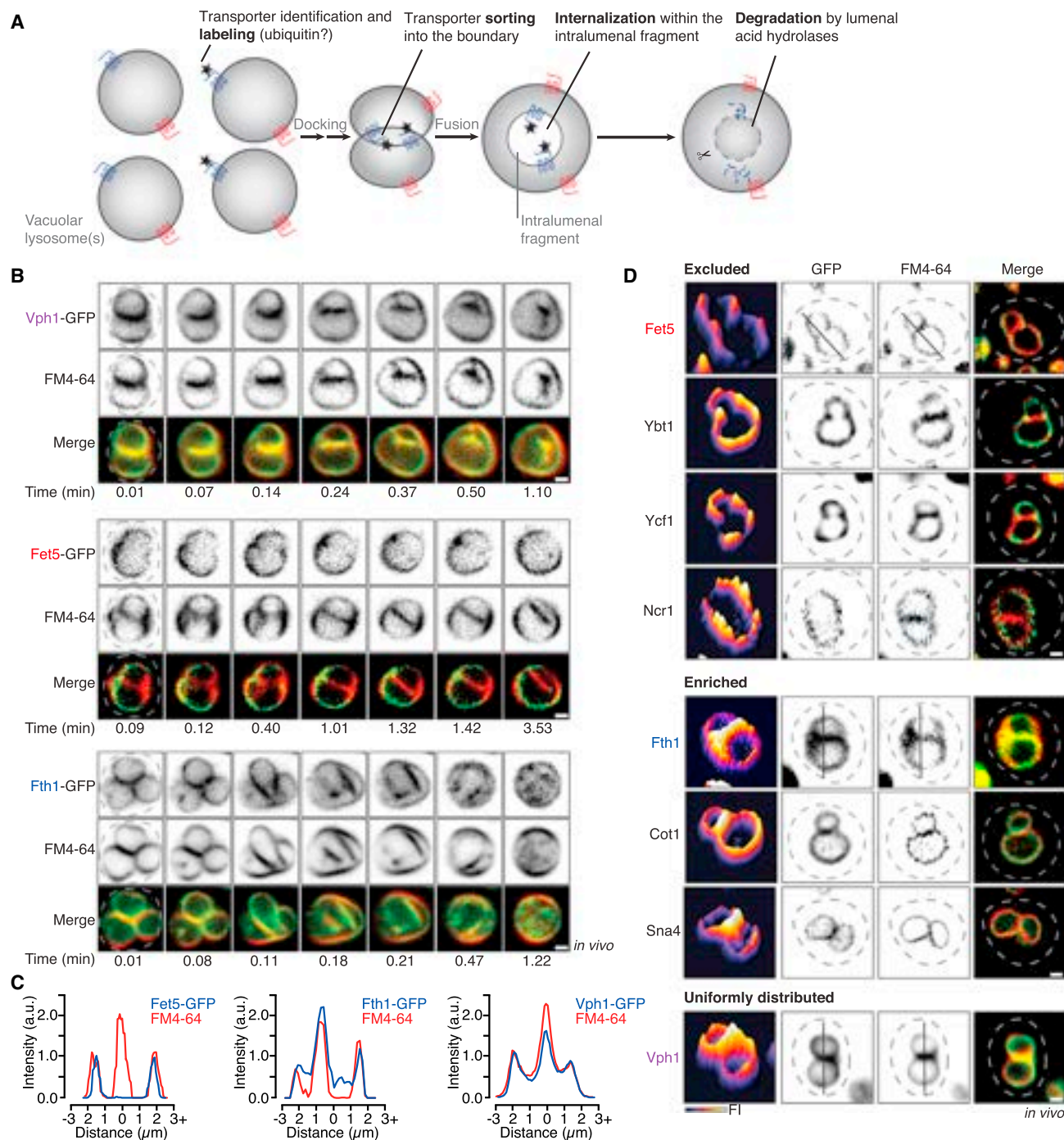


Figure 1. Polytopic Proteins Are Selectively Sorted for Degradation during Homotypic Lysosome Fusion within Live Cells

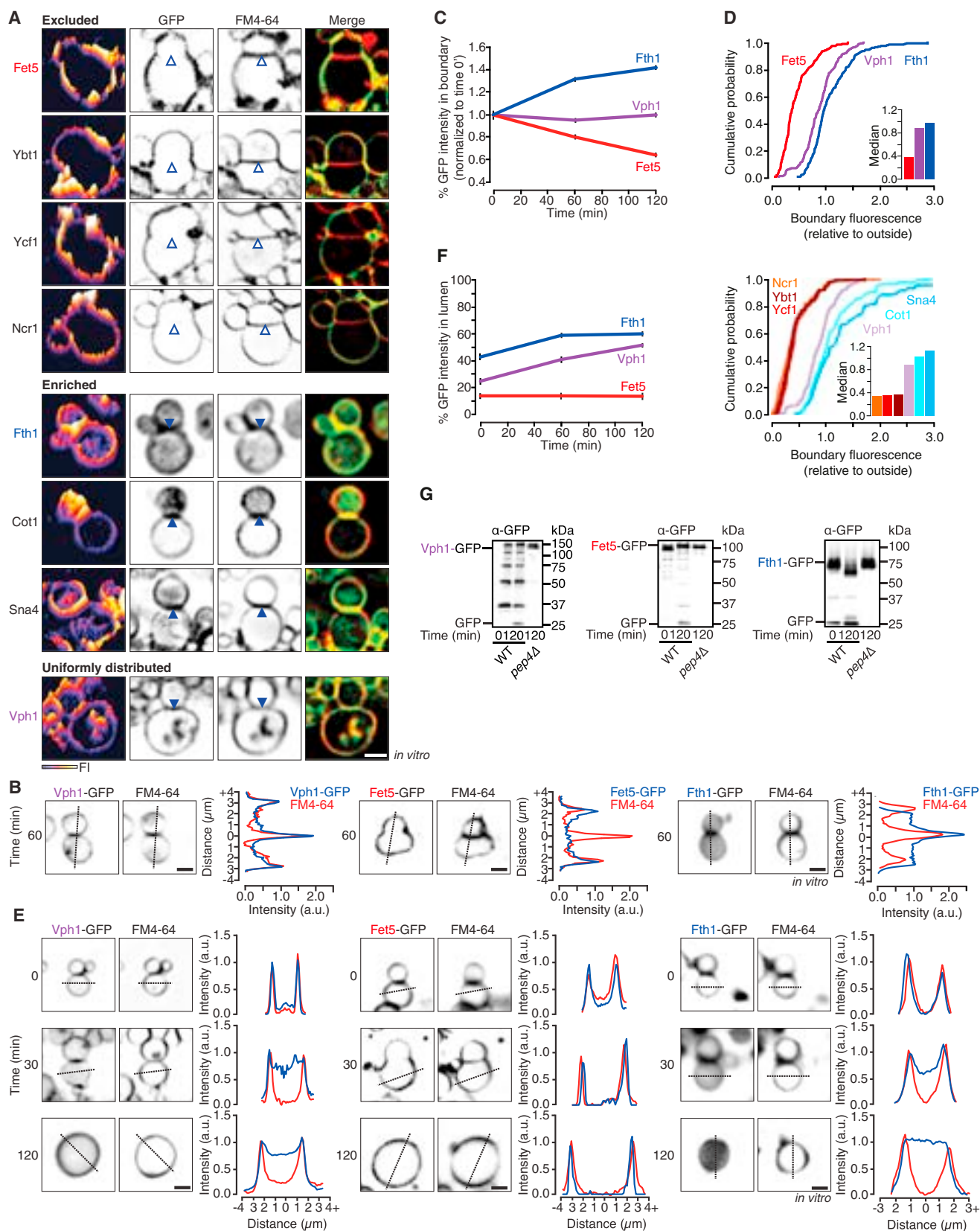
(A) Working model describing how lysosomal membrane proteins are selectively sorted into the boundary for internalization into the vacuolar lumen and subsequent degradation by the intraluminal fragment (ILF) pathway.

(B) Images from time-lapse videos of live yeast cells undergoing vacuole fusion in vivo, expressing Vph1-GFP, Fet5-GFP, or Fth1-GFP stained with FM4-64 to label the vacuolar membrane. See also [Movies S1](#), [S2](#), and [S3](#).

(C) Line-scan analysis of Fet5-GFP, Fth1-GFP, or Vph1-GFP along dotted line illustrating GFP (blue) or FM4-64 (red) fluorescence distribution (same images as shown in D).

(D) Micrographs of yeast cells expressing eight different GFP-labeled lysosomal polytopic proteins illustrating the three sorting phenotypes by the ILF pathway; excluded, enriched, or uniformly distributed. Dotted lines outline each cell as observed by DIC. GFP fluorescence intensity profile plots (left panel).

Scale bars, 1 μm .



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2014; Fang et al., 2014). Thus, we hypothesized that one function of the ILF pathway may be to clear damaged polytopic proteins from lysosome membranes. To test this hypothesis, we first treated live yeast cells with heat stress, a known trigger of protein misfolding or damage (e.g., Keener and Babst, 2013), and studied its effect on the distribution of GFP-labeled polytopic proteins during vacuole fusion (Figure 3A; Movies S4 and S5). As predicted, Fet5-GFP (an excluded protein) was shunted to the boundary and internalized within the lumen when cells were heat stressed, whereas Fth1-GFP (an enriched protein) continued to be cleared from the lysosome membrane. Indeed, all polytopic proteins tested were sorted into the boundary and internalized regardless of their sorting phenotypes (Figure 3B), suggesting that the ILF pathway functions to clear misfolded lysosomal proteins within cells.

We next repeated these experiments using isolated vacuolar lysosomes and made similar findings (Figure 3C), confirming that the molecular machinery required to sense and sort misfolded polytopic proteins copurifies with the organelles. Because heat stress could also cause misfolding of fusogenic proteins, we measured homotypic vacuole fusion using a luminal content-mixing assay (Jun and Wickner, 2007) and confirmed that heat stress does not completely block the fusion reaction in vitro, suggesting that the machinery is intact (Figure 3D). To better assess protein sorting in response to heat stress, we next used micrographs shown in Figures 2A and 3C to calculate the proportion of docked vacuoles that contained Fet5-GFP or Fth1-GFP within their boundary membranes (Figure 3E). As expected, more Fet5-GFP was sorted into the boundary upon heat stress, whereas Fth1-GFP was enriched in the boundary under both conditions. Together, these results demonstrate that misfolded polytopic proteins are sorted into the ILF pathway for clearance.

To demonstrate that damaged proteins are internalized upon fusion, we used micrographs shown in Figures 2A and 3C to calculate the proportion of total Fet5-GFP or Fth1-GFP fluorescence found within the lumen of vacuoles after fusion occurred (Figure 3F). As predicted, protein damage induced by heat stress increased luminal levels of both proteins, suggesting their internalization by the ILF pathway. We next replaced cytoplasmic GFP tags with pHluorin, a pH-sensitive variant, and monitored fluorescence (an indicator of internalization) during the course of the vacuole fusion reaction in vitro. Our reasoning was that because the luminal pH is lower (pH 5; Brett et al., 2005) than that of the reaction buffer (pH 6.8), a drop in fluorescence intensity occurs upon internalization of pHluorin-tagged proteins (Figure S1). As predicted, Fth1-pHluorin but not Fet5-pHluorin fluorescence was quenched over time under control conditions

(Figure 3G), consistent with Fet5 being spared and Fth1 being internalized during fusion. However, the fluorescence of both pHluorin-tagged proteins was quenched under heat stress, confirming that damaged polytopic proteins are internalized within the vacuole lumen during fusion.

Finally, we confirmed that these internalized proteins are degraded by western blot analysis (Figure 3H). We observed that pretreating the vacuoles with heat stress increased Fet5-GFP cleavage upon fusion, consistent with it being shunted to the boundary and internalized under these conditions, whereas heat stress had a minimal effect on Fth1-GFP cleavage as it is internalized under both conditions. GFP cleavage was abolished when *PEP4* and *PRB1* were deleted, confirming that observed proteolysis is conducted by luminal hydrolases. Taken together, these findings demonstrate that the ILF pathway is responsible for clearing misfolded lysosomal polytopic proteins from membranes and thus contributes to cellular protein quality control.

TOR Activation by Cycloheximide Triggers Protein Degradation by the ILF Pathway

Because some proteins studied were readily internalized and others were spared during organelle fusion under control conditions, we reasoned that they had different lifetimes. To better assess their rates of turnover, we employed a classic pulse-chase approach that involves treating live cells with the protein translation inhibitor cycloheximide (CHX) and then monitored protein degradation by fluorescence microscopy (Watanabe-Asano et al., 2014). We predicted that when translation is blocked, levels of proteins that are enriched within the boundary would be depleted (as proteins degraded during lysosome fusion events would not be replenished), whereas levels of excluded proteins would persist over time. However, when we treated live yeast cells with low concentrations of CHX (100 μ M for 90 min), we observed sorting of all GFP-tagged polytopic proteins studied into the ILF pathway (Figures 4A and 4B; Movies S6 and S7), including Fet5-GFP and other proteins that are normally spared. To better understand the basis of this observation, we repeated this experiment with isolated organelles that do not copurify with polysomes, ER or ribosomal components that are known targets of CHX (Schneider-Poetsch et al., 2010; Cabrera and Ungermann, 2008). Again, to our surprise, CHX caused the immediate sorting (Figures 4C and S2), internalization (Figures 4D and S2), and degradation (Figures 4E and S2) of all lysosomal polytopic proteins studied during organelle fusion in vitro. Although perplexing, these results suggest that CHX has secondary targets on lysosomes that trigger polytopic protein degradation by the ILF pathway.

Figure 2. Polytopic Proteins Are Sorted for Degradation during Lysosome Fusion In Vitro

(A) Micrographs of isolated vacuoles expressing a different GFP-labeled membrane protein stained with FM4-64 under standard fusion conditions. Boundaries containing (closed arrowheads) or lacking (open arrowheads) GFP fluorescence are indicated.
 (B) Membrane distribution of Vph1-GFP, Fet5-GFP, or Fth1-GFP on isolated vacuoles at 60 min into the fusion reaction highlighting GFP localization in the boundary, with line-scan analysis highlighting GFP fluorescence in the boundary.
 (C) Percentage of GFP fluorescence in the boundary observed over the course of the in vitro fusion reaction calculated from micrographic data ($n \geq 168$).
 (D) Cumulative probability curves of GFP fluorescence intensity within the boundary membrane (relative to outside) for all proteins tested in vitro.
 (E and F) Membrane distribution of Vph1-GFP, Fet5-GFP, or Fth1-GFP on isolated vacuoles at 0, 30, and 120 min into the fusion reaction with line-scan analysis indicating GFP fluorescence in the lumen (E). Associated GFP luminal fluorescence intensity measurements are shown in (F) ($n \geq 168$).
 (G) Western blots comparing Vph1-GFP, Fet5-GFP, and Fth1-GFP degradation kinetics in wild-type (WT) or *pep4 Δ prb1 Δ* (*pep4 Δ*) strains in vitro. Data in (C) and (E) are presented as means $\pm$ SEM. Scale bars, 2 μ m.

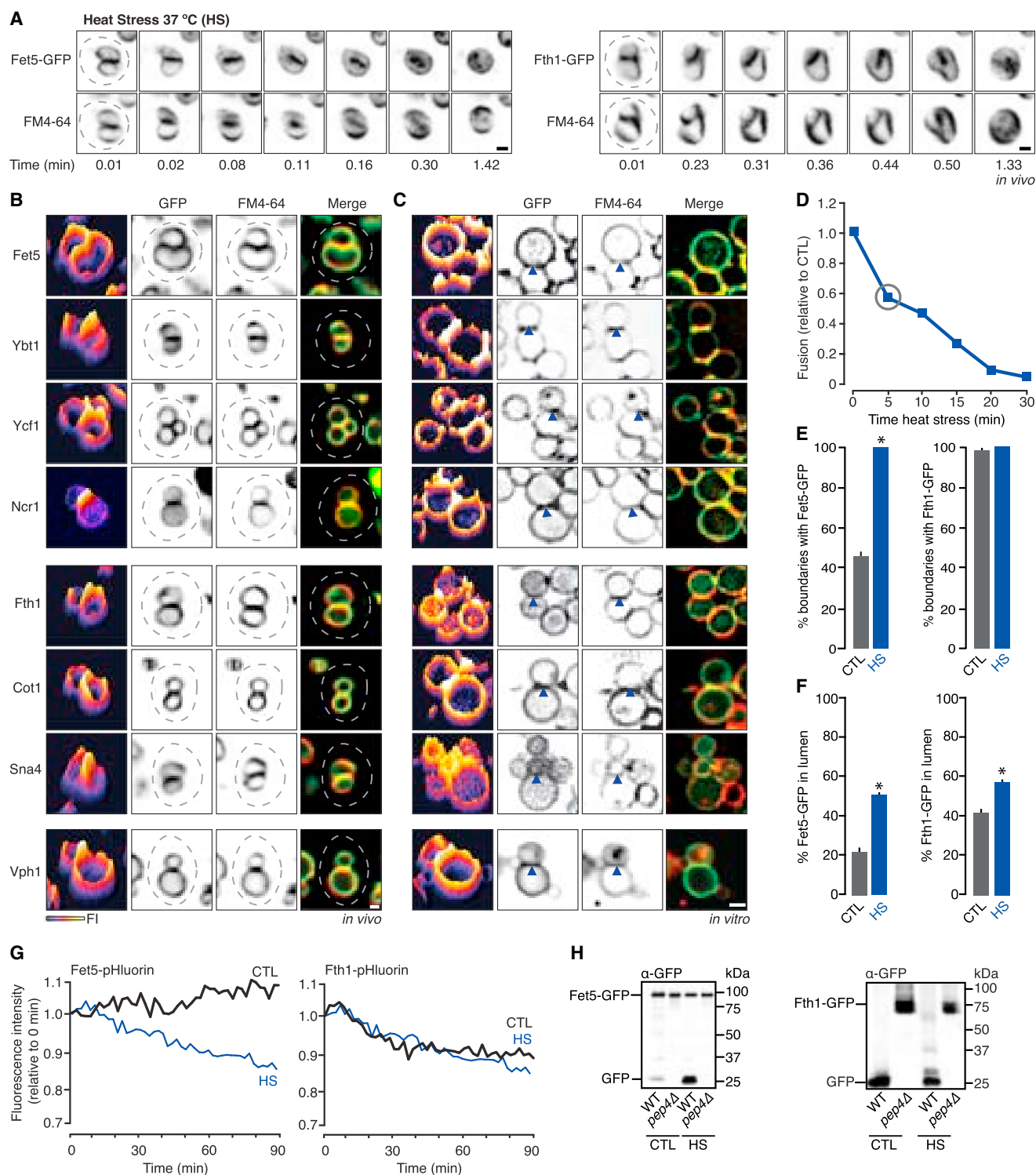


Figure 3. The ILF Pathway Degrades Misfolded Lysosomal Proteins

(A–C) Images from time-lapse videos of vacuole fusion events within cells expressing Fet5-GFP or Fth1-GFP after heat stress (A). Micrographs of docked vacuoles expressing eight different polytopic proteins tagged with GFP after heat stress were imaged *in vivo* (B) and *in vitro* (C). (D) *In vitro* homotypic vacuole fusion measured using a luminal enzyme complementation assay after isolated organelles were pretreated at 37°C for indicated times. Gray circle indicates time of heat stress used in all other *in vitro* experiments shown. (E and F) Percentage of boundaries observed containing Fet5-GFP or Fth1-GFP (E) and GFP fluorescence intensity within the lumen of docked vacuoles 60 min into the fusion reaction based on micrographic analysis (F) ($n \geq 360$). CTL, standard (control) conditions; HS, after heat stress.

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CHX is structurally similar to bioactive compounds that do not target ribosomes, and it has been reported for decades to have off-target effects (e.g., MacDonald and Ellis, 1969; McMahon, 1975). In fact, surface polytopic protein degradation is also triggered by CHX, supposedly through the TOR (target of rapamycin) signaling pathway (MacGurn et al., 2011), which releases amino acids from lysosomal stores to accommodate protein translation by ribosomes (Loewith and Hall, 2011). Although controversial, this finding may explain our observations, especially since the TOR signaling machinery functions on lysosome membranes, copurifies with vacuolar lysosomes, and is known to mediate major changes in lysosome physiology that could involve restructuring the protein content of lysosomal membranes (Settembre et al., 2013). Thus, to test whether the effects of CHX on the ILF pathway were mediated by TOR signaling, we pretreated isolated vacuolar lysosomes with rapamycin, a TOR kinase inhibitor (7 μ M for 15 min; Michailat et al., 2012), with the expectation of abolishing CHX-mediated degradation of polytopic proteins by the ILF pathway. As expected, rapamycin blocked CHX-mediated sorting (Figure 4F), internalization (Figure S2), and degradation (Figure 4G) of Fet5-GFP, a polytopic protein that is degraded by the ILF pathway upon CHX treatment but otherwise spared. Deleting FPR1, the target of rapamycin in yeast cells (Heitman et al., 1991), abolished this effect (Figures 2F and S2), confirming that the drug inhibits TOR signaling. Rapamycin alone or DMSO had no effect on Fet5-GFP sorting or degradation (Figure S2) and has no measurable effect on the membrane fusion reaction (Michailat et al., 2012). Furthermore, rapamycin did not affect sorting of Fth1-GFP, which does not require CHX for degradation as it is constitutively sorted into the ILF pathway (Figure S2). Finally, as an alternative to CHX, we treated isolated organelles with puromycin, an aminonucleoside antibiotic that targets ribosomes causing premature chain termination, at a concentration known to block protein translation (Cary et al., 2014) and found that it does not affect sorting (Figure 4F) or degradation (Figure 4H) of Fet5-GFP or Fth1-GFP (Figure S2) in the presence or absence of rapamycin. Together, these findings suggest that TOR signaling mediates polytopic protein degradation by the ILF pathway in response to CHX.

pH Regulates V-ATPase Degradation by the ILF Pathway

Surface levels of transporters and receptors must be tightly regulated for cellular physiological processes, from regulating nutrient uptake for metabolic homeostasis to properly detecting external hormonal or environmental cues (Van Belle and André, 2001). Often, substrate or ligand concentrations determine polytopic protein surface levels. For example, surface transporters Mup1, Fur4, and Can1 are downregulated and degraded by the MVB pathway upon substrate withdrawal (MacDonald et al., 2012; Keener and Babst, 2013; Ghaddar et al., 2014). Similarly, Ypq1, a lysosomal lysine transporter, is downregulated and degraded through the MVB pathway upon lysine withdrawal (Li et al., 2015a). Thus, we reasoned that the ILF pathway may con-

trol the levels of lysosomal membrane proteins in a similar manner, to mediate organellar contributions to cellular homeostasis, metabolism, or signaling.

To test this hypothesis, we studied the effects of changing $[H^+]$ on Vph1-GFP turnover by the ILF pathway (Figure 1B; Wang et al., 2002). Vph1 is the stalk domain of the V-type H^+ -ATPase complex that shunts H^+ from the cytoplasm to the lysosome lumen (Forgac, 2007). Besides providing the acid environment required for lysosomal hydrolase function, the V-ATPase is critical for cytoplasmic pH homeostasis (Diakov and Kane, 2010): The pump is inactivated when free cytoplasmic $[H^+]$ is low to prevent it from increasing pH to a level that is toxic to the cell. Conversely, it is activated to sequester H^+ within the vacuolar lumen when cytoplasmic $[H^+]$ is dangerously high. Thus, to mimic cytoplasmic acid or alkaline stress, we shifted the pH of the fusion reaction buffer from 6.80 to 6.38 or 7.32, respectively, and examined the effects on Vph1-GFP sorting (Figures 5A and 5B), internalization (Figure 5C), and degradation (Figure 5D) by the ILF pathway *in vitro*. Under acid stress, when substrate levels are high, we find less Vph1-GFP in boundary membranes, less luminal GFP fluorescence, and less GFP cleavage, demonstrating that Vph1 is spared under conditions that require V-ATPase activation for cell survival. In contrast, under alkali stress, we find enrichment of Vph1-GFP in the boundary membrane, an increase in luminal GFP fluorescence, and more GFP cleavage, demonstrating that Vph1 is downregulated under conditions that promote V-ATPase inactivation.

However, fusing GFP to Vph1 could change its lifetime or account for the observed changes in its turnover in response to acid or alkali stress. To eliminate these possibilities, we studied endogenous Vph1 degradation under pH stress by western blot analysis (Figure 5E) and found a similar pattern of degradation as observed for Vph1-GFP (Figure 5D), confirming that the epitope tag had no effect on Vph1 turnover. It is also possible that our observations were a consequence of simply altering the rates of organelle fusion. For example, acid stress could inhibit fusion, accounting for the observed decrease in internalization and degradation of Vph1-GFP. To eliminate this possibility, we measured homotypic vacuole fusion using a content-mixing assay (Figure 5F) and confirmed that acid and alkali stress had no impact on organelle fusion *in vitro*. pH could also indiscriminately affect the machinery responsible for sorting within the ILF pathway. If so, we expect that sorting of other polytopic proteins cleared by the ILF pathway should also be affected by acid and alkali stress. However, when we studied GFP-labeled Fth1 (Figures 5G–5J) or Fet5 (Figure S3; neither translocate H^+), we found that sorting, internalization, and degradation of both proteins were not affected by these stresses. Together, these results demonstrate that the ILF pathway selectively regulates membrane expression of Vph1, and presumably the activity of the V-ATPase, in response to changes in $[H^+]$, implicating a role in cellular pH homeostasis and confirming that it can mediate substrate-dependent protein degradation.

(G) Relative pHluorin fluorescence of isolated vacuoles expressing Fet5-pHluorin or Fth1-pHluorin under control conditions (CTL) or after heat stress (HS).

(H) Western blot analysis of Fet5-GFP or Fth1-GFP degradation after vacuole isolated from WT or *pep4 Δ prb1 Δ* cells were permitted to fuse for 120 min under control conditions (CTL) or after heat stress (HS).

Data are presented as means $\pm$ SEM. * $p < 0.05$. Scale bars, 1 μ m (in vivo) and 2 μ m (in vitro). See also Figure S1; Movies S4 and S5.

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Protein Sorting in the ILF Pathway Does Not Require Autophagy or ESCRT Machinery

Findings presented thus far answer the first question posed earlier by demonstrating that the ILF pathway functions to control lysosome polytopic protein quality and quantity for organelle homeostasis and for membrane protein remodeling to accommodate cellular metabolism or ion homeostasis. But what mechanisms underlie this process? Given that lysosomes contribute to autophagy and the MVB pathway, we first examined whether the ILF pathway shares underlying molecular machinery with these two major cellular protein degradation pathways. Although there is no evidence of autophagosomes or MVBs contributing to lysosomal polytopic protein sorting or internalization (Figures 1, 2, 3, 4, and 5), it remains possible that mechanisms responsible for protein sorting in these pathways may also function in the ILF pathway, as they have been implicated in the degradation of lysosome transporters and can be found on lysosome membranes (Li et al., 2015a, 2015b). To eliminate this possibility, we deleted genes encoding essential components for autophagy, *ATG1* and *ATG7*, or for the MVB pathway, *VPS27* (ESCRT-0), *VPS23* (ESCRT-I), and *VPS36* (ESCRT-II), and examined effects on GFP-tagged protein distribution during homotypic vacuole fusion events (Figures S4 and S5). Deleting any of these genes had no effect on protein sorting in vivo (Figures S4A and S4B) or in vitro (Figures S5A and S5B) or on degradation (Figure S5D) even under conditions that induced protein clearance by the ILF pathway (Figures S4C, S4D, S5C, and S5D). Thus, we are confident that the ILF pathway uses separate machinery from the autophagic and MVB pathways, representing a novel mechanism for cellular protein degradation.

The Docking Machinery Sorts Polytopic Proteins for Degradation

From the micrographic evidence presented, it is apparent that protein sorting occurs during the docking stage of the homotypic vacuolar lysosome fusion reaction when the boundary membrane is formed (Wang et al., 2002; Wickner, 2010). During docking, fusogenic lipids and proteins accumulate at the initial site of contact between organelles, where they arrange into an expanding ring at the vertex between apposing membranes. This vertex ring represents the border between outer and boundary areas of the lysosome membrane where polytopic proteins are sorted to be spared or degraded. Thus, we hypothesized that the machinery assembled at the vertex ring may contribute to polytopic protein sorting, as well as driving membrane fusion. To test this hypothesis, we blocked stages of the in vitro fusion reaction with protein or chemical inhibitors (Mayer and Wickner, 1997; Wickner and Haas, 2000; Wang et al., 2002) and monitored

effects on the membrane distribution of Fet5-GFP or Fth1-GFP (Figure 6). We first targeted the lysosomal Rab-GTPase Ypt7 that, when activated, orchestrates recruitment and assembly of the fusion machinery at the vertex ring (Brett et al., 2008). We reasoned that inactivating Ypt7 by adding rGyp1-46 (purified recombinant active domain of the Rab-GAP protein Gyp1; Rak et al., 2000) and rGdi1 (a Rab chaperone protein that extracts Ypt7 from vacuole membranes; e.g., Brett et al., 2008) or rGdi1 alone to fusion reactions should disrupt protein sorting (Eitzen et al., 2000). We found that these Ypt7 inhibitors prevented sorting of Fet5-GFP or Fth1-GFP into the boundary membrane (Figures 6A and 6B), even when protein clearance was triggered with heat stress (Figures 6C and 6D) or CHX (Figures S6A and S6B), suggesting that active Ypt7 is important for selective entry, but not exclusion, of polytopic proteins into the boundary membrane during organelle fusion. Likewise, internalization (Figures 6E and S6C) and degradation (Figure 6F) of both polytopic proteins were blocked by Ypt7 inhibitors, illustrating the importance of the fusion machinery for lipid bilayer merger required to internalize polytopic proteins for degradation by the ILF pathway.

Although this result implies a role for the docking machinery in protein sorting, an alternative interpretation is that blocking the membrane fusion reaction indirectly impairs protein sorting into the ILF pathway. Thus, we sought an approach to disrupt Ypt7 function without blocking membrane fusion to better understand its role in protein sorting. This involved using the recombinant soluble Qc-SNARE protein Vam7 (rVam7) to drive the lysosome fusion reaction, as it bypasses the need for Rab activity (Thorngren et al., 2004). In the presence of rVam7, both Fet5-GFP and Fth1-GFP are present in the boundary membrane (Figures 6A and 6B) and internalized (Figure 6E) in the absence or presence of heat stress (Figures 6C–6E) or CHX (Figure S6). Thus, bypassing Ypt7 activity blocks exclusion, but not inclusion, of polytopic proteins into the boundary membrane during fusion. Together, these results suggest that active Rab must be engaged to selectively sort polytopic proteins into the ILF pathway.

Protein Sorting Occurs during Vertex Ring Expansion

All things considered, it seems that the docking machinery contributes to polytopic protein sorting at the vertex ring that surrounds the boundary membrane during lysosome fusion. If so, this machinery could function by employing a protein sieving mechanism that sorts polytopic proteins encountered within the lysosome membrane as the vertex ring expands. Alternatively, it could recognize and actively move polytopic proteins marked for degradation across the vertex ring into the boundary after docking is completed (Figure 7A). To test these models, we first examined the membrane distribution of Fet5-GFP during

Figure 4. The ILF Pathway Degrades Proteins in Response to TOR Signaling Triggered by Cycloheximide

(A–C) Images from time-lapse videos of vacuole fusion events within yeast cells expressing Fet5-GFP or Fth1-GFP after incubation with cycloheximide (CHX) (A). Micrographs of docked vacuoles expressing different polytopic proteins tagged with GFP after CHX treatment were imaged within cells (B) or in vitro (C). (D) Relative fluorescence of isolated vacuoles expressing Fet5-pHluorin or Fth1-pHluorin during the in vitro fusion reaction under control conditions (CTL) or after CHX treatment. (E) Western blot analysis of Fet5-GFP or Fth1-GFP degradation after vacuoles isolated from WT or *pep4Δprb1Δ* cells were permitted to fuse for 120 min under CTL or CHX treatment. (F) Micrographs of docked vacuoles expressing Fet5-GFP or Fet5-GFP:*fpr1Δ* in the absence or presence of rapamycin (RAP), CHX, or puromycin (PUR). (G and H) Western blot analysis of Fet5-GFP degradation under RAP, CHX (G), and (H) PUR treatment. Scale bars, 1 μ m (in vivo) and 2 μ m (in vitro). See also Figures S1 and S2; Movies S6 and S7.

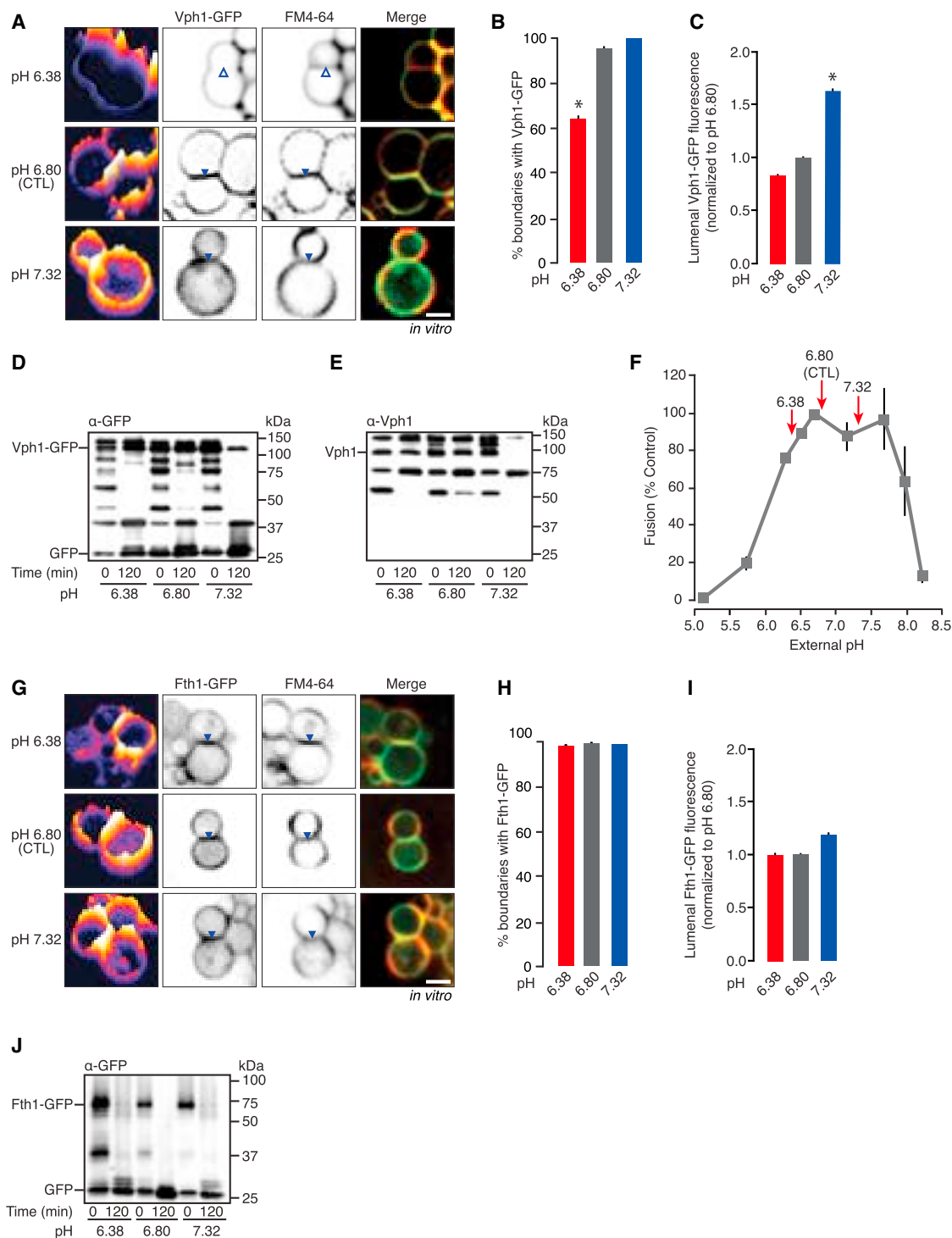


Figure 5. Changing pH Affects Degradation of Vph1-GFP by the ILF Pathway

(A) Micrographs of docked vacuoles expressing Vph1-GFP imaged 60 min into the *in vitro* fusion reaction titrated to different pH values. (B–E) Percentage of boundaries containing Vph1-GFP (B) and relative luminal GFP fluorescence (C) (normalized to pH 6.80; standard conditions) were calculated from micrographic data ($n \geq 308$). Western blots stained with antibodies against GFP (D) or Vph1 (E) indicating degradation kinetics of Vph1-GFP (α-Vph1 was not against Vph1-GFP) before and after fusion of isolated vacuoles at different pH values. (F) Homotypic vacuole fusion of isolated vacuoles was measured after reactions were titrated over a range of pH values. Arrows indicate experimental pH values used in other experiments shown ($n = 3$).

(G) Micrographs of docked vacuoles expressing Fth1-GFP imaged at 60 min into the *in vitro* fusion reaction titrated to three pH values.

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early stages of the *in vitro* fusion reaction when docking occurs (Jun et al., 2006). For accurate study of early docking events, it is important to consider that some docked interfaces observed at early time points could be mature because organelle interactions can occur prior to triggering fusion, either within yeast cells (which contain 2–3 vacuoles) prior to isolation or during the organelle isolation process. Thus, to distinguish newly formed organelle interactions from existing mature contact sites, we mixed vacuolar lysosomes isolated separately from two different yeast strains, one expressing Fet5-GFP and the other lacking a GFP tag, stained their membranes with FM4-64 to visualize contact sites, and analyzed only boundary membrane between Fet5-GFP-labeled and GFP-free organelles (Figure 7B). To assess vertex ring assembly at these early time points, we also monitored the membrane distribution of Vps33-GFP using the same approach (Figure 7C). We selected Vps33 because it is one of six subunits that make up the HOPS tethering holocomplex, a key component of the fusion machinery that is recruited to the vertex ring by active Ytp7 as part of the docking subreaction (Wang et al., 2002). Hence, enrichment of Vps33-GFP at the edges of the boundary membrane between GFP-labeled and GFP-free vacuoles labels fusion machinery assemblies at the vertex ring triggered by ATP *in vitro*. By measuring boundary lengths over time, we confirmed that docking occurred at the same rate and terminated at the same time (30 min) in both samples (Figure 7E), consistent with previous reports (e.g., Wang et al., 2002). Importantly, Fet5-GFP exclusion from boundaries occurred at the same rate as vertex ring assembly, assessed by monitoring Vps33-GFP enrichment at organelle vertices over time (Figure 7F), whereby the half-times of boundary expansion, Vps33-GFP enrichment, and Fet5-GFP exclusion were similar, strongly suggesting that polytopic protein exclusion from the boundary occurs during vertex ring expansion.

But does the same model also describe sorting of proteins into the boundary for degradation? To address this question, we used fluorescence recovery after photobleaching (FRAP) to study lateral mobility of Fth1-GFP (an enriched protein) within the vacuole membrane during early or late stages of the docking subreaction *in vitro*. We hypothesized that if sorting occurred during vertex ring expansion, then Fth1-GFP should be mobilized into the boundary early (10 min) but not late into the reaction (60 min) after the fusion machinery is stabilized and no further increase in the boundary length is observed (Figure 7E). As predicted, Fth1-GFP fluorescence recovered to a significantly higher level after photobleaching within the boundary membrane at 10 min (42.9%) compared with 60 min (17.9%; Figures 7G and 7H). Fth1-GFP mobility on the outer area of the vacuole membrane did not change over the course of the fusion reaction, and values were similar to those obtained when studying the boundary membrane at 60 min. These values were also similar to Vph1-GFP mobility on the outer and boundary membranes at early and late times (Figures 7I and 7J). Because Vph1-GFP is uniformly distributed on the vacuole membrane (Figures 1, 2,

and 5; Wang et al., 2002), we speculate that these values (mobile fractions between 10.8% and 14%) represent a good measure of polytopic protein lateral mobility within the vacuolar lysosome membrane. We also assessed lipid mobility by recording FM4-64 fluorescence recovery (Figure S7). As expected, the mobile fraction of FM4-64 within the lipid bilayer was much higher (80%; as previously reported in Dhonukshe et al., 2006) than the polytopic proteins studied, lending confidence to our assessment of Fth1-GFP movement during vacuole fusion. All things considered, the lateral mobility of Fth1-GFP is highest within the boundary membrane during the early stage of docking, lending support to a molecular sieving model of protein sorting into the ILF pathway that occurs during vertex expansion.

DISCUSSION

The ILF Pathway Is a New Mechanism Responsible for Organelle Polytopic Protein Degradation within Cells

Lysosomal polytopic proteins are critical for cellular nutrient recycling, signaling, and ion homeostasis. Thus, it is important that we understand how they are regulated or degraded. Here we describe a new pathway responsible for turnover of transporters and membrane proteins on lysosomes. Using fluorescence microscopy, we show that GFP-tagged polytopic proteins are selectively sorted into the boundary area of apposing membranes that is internalized into the lumen upon lipid bilayer fusion within living cells (Figures 1, 3, 4, and S4). We confirmed our original observations by monitoring protein internalization using a pHluorin-based fluorescence assay and show by western blot that internalized proteins are degraded by luminal hydrolases (Figures 2, 3, 4, 5, 6, and S6). Of the eight studied, four proteins were spared and four were internalized by this pathway under normal growth conditions, indicating that lysosomal polytopic proteins have different lifetimes and that the protein content of the lysosomal membrane can change with every homotypic fusion event. Although it is not clear why the cell would want to rapidly turn over some proteins (e.g., Fth1) and not others (Fet5), this observation is consistent with reports of surface membrane proteins that are also constitutively degraded; for example, the G-protein-coupled receptor Ste3 is constitutively internalized and degraded by the MVB pathway (Davis et al., 1993). Because some proteins were spared, it was possible that they utilized an alternative mechanism for degradation, such as the vRED pathway (Li et al., 2015a, 2015b). However, we found that all proteins were shunted to the ILF pathway after misfolding was induced by heat stress (Figure 3), confirming that all proteins studied can be degraded by this new pathway. Furthermore, by piggybacking membrane fusion, the ILF pathway seems like a more efficient alternative for lysosomes to turn over their membrane proteins compared with the vRED pathway, which requires additional trafficking steps to deliver lysosomal proteins to the MVB for packaging and back to the lysosome for degradation (Li et al., 2015a). Although lysosomes

(H and I) Percentage of boundaries containing Fth1-GFP (H) and relative luminal GFP fluorescence (I) (normalized to pH 6.80) were calculated from micrographic data ($n \geq 91$).

(J) Western blot kinetics of Fth1-GFP degradation before and after fusion of isolated vacuoles at different pH values.

Data are presented as means $\pm$ SEM. * $p < 0.05$. Scale bars, 2 μ m. See also Figure S3.

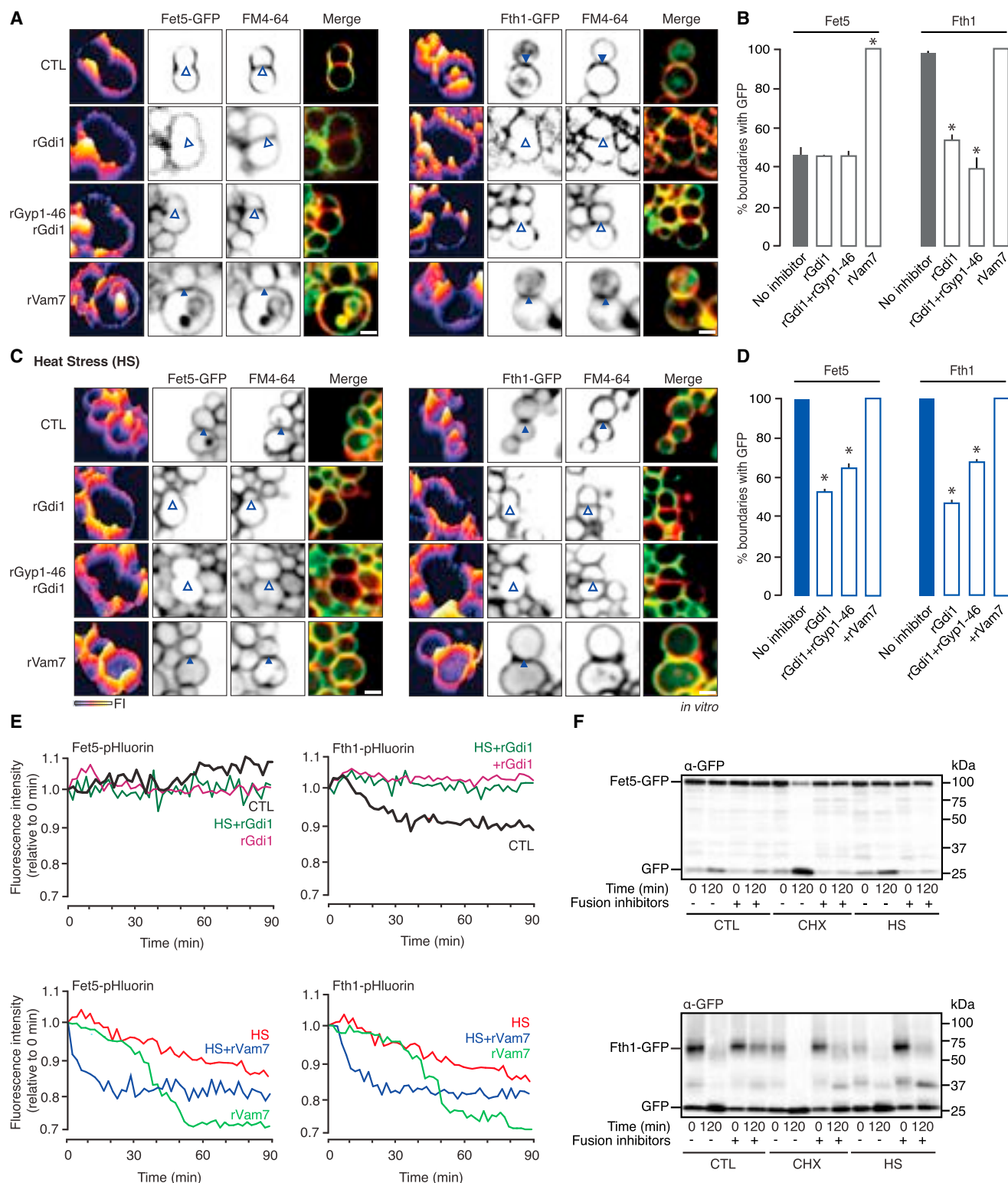


Figure 6. Lysosomal Polytopic Protein Sorting and Internalization Requires the Membrane Fusion Machinery

(A) Micrographs of docked vacuoles expressing Fet5-GFP or Fth1-GFP during the *in vitro* fusion reaction in the absence (CTL) or presence of 4 μ M rGdi, 4 μ M rGdi, and 3.2 μ M rGyp1-46, or 100 nM rVam7.

(B) Percentage of boundaries containing Fet5-GFP or Fth1-GFP under these conditions calculated from micrographic data ($n \geq 103$).

(C) Micrographs of docked vacuoles expressing Fet5-GFP or Fth1-GFP *in vitro* after heat stress.

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frequently undergo homotypic fusion (Wickner, 2010), they require heterotypic fusion events between MVBs or autophagosomes for delivery of incoming biomaterials to be recycled (Carlsson and Simonsen, 2015). As similar fusion machinery mediates these events (Luzio et al., 2009; Jiang et al., 2014), it is possible that intraluminal membrane fragments also form. Current advances in light and electron microscopy will allow us to test this hypothesis in the near future. In the meantime, it is tempting to speculate that the ILF pathway could allow these dynamic organelles to remodel the protein content within their membranes while they receive biomaterials for recycling.

But how are intraluminal fragments degraded? To prevent lysosome rupture, membrane lipids and proteins facing the lumen are resistant to attack by hydrolases (see Kolter and Sandhoff, 2005). However, when a portion of this membrane is internalized through the ILF pathway, the membrane conserves its orientation (i.e., the luminal face remains luminal) but becomes susceptible to degradation. Although many factors may be at play (e.g., perhaps selective lipid sorting into the boundary membrane promotes rupture or susceptibility to hydrolases; see Fratti et al., 2004), we speculate that this may be an outcome of lipid bilayer merger driven by asymmetrically aligned SNARE complexes (D'Agostino et al., 2016): SNARE domains of *trans* complexes extend into the cytoplasm around the vertex ring, where they interact with the fusion machinery (e.g., HOPS; Wickner, 2010). Upon zippering, they form *cis* complexes that exclusively reside in the limiting membrane of the fusion product, not within the intraluminal fragment. As transmembrane domains of SNARE complexes seem to stabilize the lipid bilayer during merger, the absence of these complexes within the fragment may destabilize the bilayer, exposing the inner leaflet to luminal hydrolases to initiate degradation. We have begun to characterize intermediates of this lipid bilayer fusion reaction in the context of intraluminal fragment formation (Mattie et al., 2016) and plan to test this hypothesis in the near future.

A Molecular Sieving Mechanism May Be Responsible for Protein Sorting in the ILF Pathway

Upon the discovery of this new protein degradation pathway, we immediately sought to better understand the molecular basis of how it works. We present many lines of evidence to support a molecular sieving model of protein sorting that occurs during the docking stage of the fusion reaction when the machinery assembles at an expanding vertex ring between apposing organelles prior to lipid bilayer merger. (1) We find that polytopic proteins are selectively sorted into or out of the boundary area of membrane formed between apposing organelles during docking (Figures 2, 7, and S7). (2) When we block or bypass the activity of Ypt7, an important mediator of docking, protein sorting is disrupted (Figures 6 and S6). (3) Using FRAP and time-lapse microscopy, we demonstrate that protein sorting occurs during expansion of the vertex ring that encircles

the boundary between apposing lysosomes (Figure 7). Topologically, this mechanism is analogous to polytopic protein sorting into budding intraluminal vesicles on the outer membrane of immature MVBs, where the ESCRT machinery assembles into a ring at the neck of the budding vesicle (Nickerson et al., 2007). However, we demonstrate that the ESCRT machinery, as well as the autophagy machinery, does not contribute to the ILF pathway (Figures S4 and S5). Furthermore, other cytoplasmic factors (e.g., components of the proteasome machinery) likely do not play a role because similar results were obtained from experiments conducted *in vivo* and *in vitro* (Figure 2). So what is the molecular basis of protein sorting in the ILF pathway?

Although we implicate molecular sieving in this process, we did not uncover the basis of protein selection. However, we are confident that it is not simply protein size, for example, as it does not correlate with the observed sorting phenotypes. Rather, we speculate that the ILF pathway uses a ubiquitin-based mechanism similar to the other protein degradation pathways in the cell, based on the following observations. Proteomic screens reveal that most of the lysosomal polytopic proteins studied can be ubiquitinated (Peng et al., 2003). Protein ubiquitylation machinery, such as adaptor proteins (e.g., Ssh4) and E3 ligases (Rsp5), are found on vacuole membranes (Pokrzywa et al., 2009; Li et al., 2015a). This includes a paralog of Sna3, an important adapter protein in the MVB pathway, called Sna4, which exclusively resides on vacuole lysosome membranes. At MVBs, Sna3 binds internalized surface polytopic proteins to accommodate ubiquitylation by Rsp5 and sorting into ILVs, and by doing so it too is internalized and degraded by the MVB pathway (MacDonald et al., 2012). Like Sna3, we show that Sna4 is sorted into the boundary of docked lysosomes and is internalized upon fusion (Figures 1 and 2), suggesting that perhaps Sna4 performs a similar function in the ILF pathway.

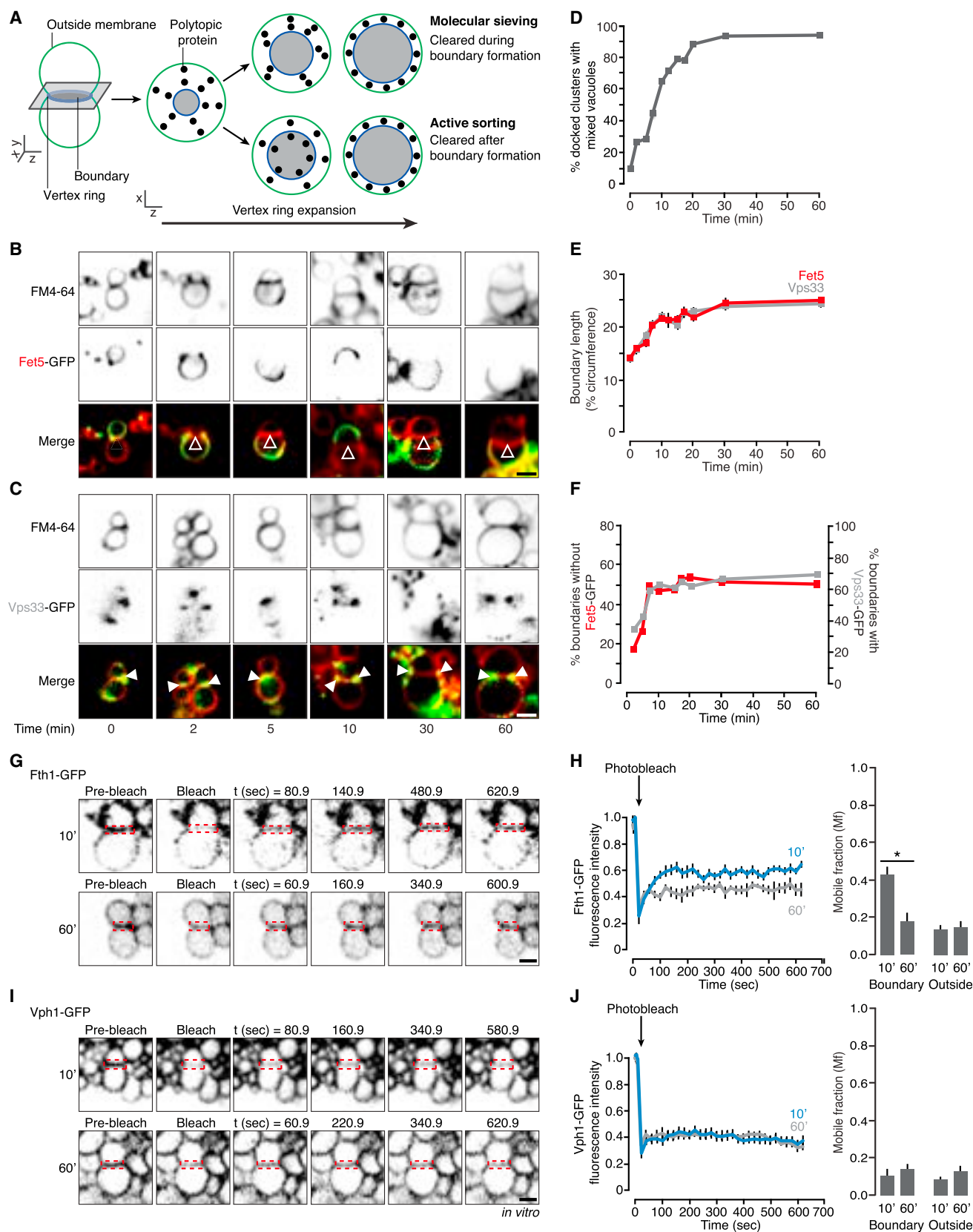
If ubiquitylation mediates degradation, labeled proteins must be recognized at the vertex ring for sorting into the ILF pathway. Of the proteins that concentrate at this site, three contain partial potential ubiquitin binding domains (UBDs): Vps11, Vps18, and Vps39 (Nickerson et al., 2009). Together with three other subunits, these proteins form HOPS, the lysosomal multisubunit tethering complex important for MVB-lysosome and homotypic lysosome fusion (Balderhaar and Ungermann, 2013). Although speculative, these three partial domains may form a UBD when in complex that recognizes and sorts ubiquitinated lysosomal polytopic proteins into the boundary, similar to how the multiple UBDs of the ESCRT machinery sort surface proteins in the MVB pathway (Shields and Piper, 2011). Using the comprehensive set of approaches to study the ILF pathway presented herein, we have begun to test this model, and it will be interesting to determine whether the ILF pathway shares protein labeling machinery but has unique sorting machinery in comparison with other protein degradation pathways.

(D) Percentage of boundaries containing Fet5-GFP or Fth1-GFP calculated from micrographic analysis ($n \geq 242$).

(E) Relative fluorescence of isolated vacuoles expressing Fet5-pHluorin or Fth1-pHluorin during *in vitro* fusion reactions incubated with rGdi or rVam7 under control (CTL) and heat shock (HS) treatment.

(F) Western blot analysis of Fet5-GFP and Fth1-GFP degradation before and after fusion of isolated vacuoles under CTL, CHX, or HS conditions.

Data are presented as means $\pm$ SEM. * $p < 0.05$. Scale bars, 2 μ m. See also Figures S4–S6.



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Importance of the ILF Pathway to Cell Physiology

What is the purpose of the ILF pathway? Here, we show that it performs at least three important functions: First, it mediates lysosomal polytopic protein degradation upon misfolding by heat stress (Figure 3). The ERAD pathway performs this function to clear damaged or unfolded proteins at the ER (Vembar and Brodsky, 2008), the proteasome for cytoplasmic proteins (Fang et al., 2014), and the MVB for surface polytopic proteins (Keener and Babst, 2013). Thus, we propose that the ILF pathway is an equally important contributor to cellular protein quality control. Second, the ILF pathway responds to changes in substrate levels to selectively regulate cognate transporter levels, as increasing pH promotes degradation of Vph1 by the ILF pathway (Figure 5), consistent with reports of how the V-ATPase regulates cellular pH homeostasis (Tarsio et al., 2011). Similar mechanisms are in place at the plasma membrane to regulate cellular nutrient uptake, receptor signaling, and osmotic homeostasis (Keener and Babst, 2013). We speculate that the ILF pathway performs an analogous function on lysosomes to potentially regulate Ca^{2+} signaling or nutrient export, for example, which are important contributors to cellular physiology and metabolism.

Lastly, the ILF pathway degrades proteins in response to TOR activation induced by the protein translation inhibitor CHX (Figure 4). Lysosomes are an important source of cellular amino acids needed to feed the protein translation machinery. Thus, the activities of these two fundamental processes are tightly regulated through extensive metabolic signaling circuitry including the TOR pathway (Efeyan et al., 2012; Settembre et al., 2013). For example, when cellular amino acid pools are depleted, translation stops, TOR is activated, and lysosome activity is enhanced to increase amino acid levels, closing a negative feedback loop. Blocking protein translation with puromycin or CHX triggers protein degradation by the ILF pathway within live cells (Figures 4A, 4B, and S2), lending support to this model. However, only CHX elicits this response when added to purified organelle preparations devoid of ribosomal components, which is abolished by rapamycin, a TOR signaling inhibitor, suggesting that it may also have a second target present on lysosome membranes that activates TOR signaling and triggers degradation of lysosomal polytopic proteins to increase cellular amino acid levels. In any case, our findings clearly demonstrate a role for the ILF pathway in cellular amino acid metabolism. Maintenance of amino acid levels through sustained or enhanced lysosome function is critical

for survival as cells age (Aris et al., 2012). As such, impaired lysosome activity correlates with many age-related disorders such as Alzheimer's disease (Settembre et al., 2013). The transcriptional regulator TFEB (transcription factor EB) plays a critical role in this process by increasing lysosome biogenesis (Settembre et al., 2013). By remodeling existing lysosomes to accommodate changes in cell physiology as they age, we argue that the ILF pathway may play a complementary and equally important role in cellular longevity.

EXPERIMENTAL PROCEDURES

Vacuole Isolation and Homotypic Vacuole Fusion

Vacuoles were isolated from yeast cells as previously described (Haas, 1995). Fusion reactions were prepared using 6 μg of vacuoles isolated from GFP derivative strains in standard fusion reaction buffer with 0.125 M KCl, 5 mM MgCl_2 , 1 mM ATP, and 10 μM coenzyme A. Vacuolar membranes were stained with FM4-64 by treating vacuoles with 3 μM FM4-64 for 10 min at 27°C. Reactions were incubated at 27°C for 60 min, unless otherwise noted, and placed on ice prior to visualization by microscopy. Where indicated, vacuoles were incubated in the absence (CTL) or presence of 3.2 μM Gyp1-46, 4 μM rGdi, or 100 nM rVam7 (in the absence of ATP). For heat stress treatment, vacuoles were pretreated for 5 min at 37°C before addition to the fusion reaction and incubation for 30 min at 27°C. Where indicated, vacuoles were pretreated with DMSO, 7 μM rapamycin, 100 μM CHX, or 100 μM puromycin for 15 min at 27°C and incubated for an additional 15 min with the fusion reaction.

For assessment of protein exclusion during vertex expansion, isolated vacuoles were stained with FM4-64 and reactions prepared by mixing vacuole populations, 3 μg of either Vps33-GFP or Fet5-GFP with 3 μg of DKY6281, and were incubated for 0–60 min.

Fluorescence Recovery after Photobleaching

FRAP experiments were conducted using vacuoles isolated from yeast strains expressing Fth1-GFP, Vph1-GFP, and Vps33-GFP, or lacking a GFP tag (DKY6281 stained with FM4-64). Fusion reactions were incubated at 27°C for 10 min or 60 min prior to imaging using a Nikon Eclipse TiE inverted microscope equipped with a Laser Scanning C2 Confocal system and a CFI ApoTIRF 1.49 numerical aperture 100 $\times$ objective lens. Regions of interest (ROIs) were first scanned using a 488-nm laser (3%–5% power) for five cycles to establish prebleached fluorescence intensity. ROIs were then photobleached for three cycles using 100% 488-nm laser power. ROI fluorescence intensity was then recorded every 20 s for 10 min at 3%–5% laser power (488-nm laser, dichroic 525/50-nm 561LP emission filter). FM4-64 FRAP experiments were conducted as described except with a dichroic 447/60-nm emission filter. Unprocessed data obtained as mean fluorescence values were corrected with respect to background fluorescence intensity and spontaneous photobleaching and then were normalized with respect to fluorescence intensity of prebleached images. Mobile fraction (Mf) values were

Figure 7. Protein Sorting Occurs during Vertex Ring Expansion

- (A) Working model demonstrating two hypotheses of polytopic protein sorting from the boundary membrane either through a molecular sieving mechanism or through active sorting.
- (B and C) Micrographs of docked vacuoles between Fet5-GFP and GFP-free organelles (B) or between Vps33-GFP and GFP-free organelles (C) acquired at different times during the *in vitro* fusion reaction.
- (D) Percentage of docked clusters that were mixed between Fet5-GFP and GFP-free organelles. A total of 2,179 docked vacuole clusters were observed.
- (E and F) Micrographic data were used to calculate the boundary membrane length between apposed vacuoles as a percentage of the circumference (E) and the percentage of boundaries either without Fet5-GFP or with Vps33-GFP (F) ($n \geq 108$).
- (G) Micrographs of FRAP analysis of the boundary membrane between docked vacuoles expressing Fth1-GFP 10 and 60 min after vacuoles were permitted to fuse.
- (H) GFP fluorescence intensity in the boundary was measured over the course of the experiment and the mobile fraction (Mf) was calculated for ROIs photobleached in the boundary or outside membranes of docked vacuoles ($n \geq 11$).
- (I) FRAP analysis of the boundary membrane between docked vacuoles expressing Vph1-GFP 10 and 60 min into the *in vitro* fusion reaction.
- (J) GFP fluorescence intensity in the boundary was measured, and Mf values for ROIs photobleached in the boundary and outside membranes were calculated from micrographic analysis ($n \geq 13$).

The red dashed box outlines ROIs subjected to photobleaching. Data are presented as means $\pm$ SEM. * $p < 0.05$. Scale bars, 2 μm . See also Figure S7.

calculated by fitting experimental values with the equation: $Mf = (F_{\text{end}} - F_{\text{post}}) / (F_{\text{pre}} - F_{\text{post}})$, where F_{pre} corresponds to prebleach fluorescence, F_{post} to fluorescence intensity after photobleaching, and F_{end} to recovered fluorescence intensity. Fluorescence intensity curves and Mf values are presented as means $\pm$ SEM.

pHluorin-Based Assay

Eclectic pHluorin was cytoplasmically tagged to lysosomal membrane proteins (Prosser et al., 2010). Fusion reactions (15 μ L) were prepared using 6 μ g of isolated vacuoles and standard reaction buffer. Reactions were transferred to black 96-well conical-bottom microtiter plates with 15 μ L of reaction buffer titrated to different pH values. pHluorin fluorescence (λ_{ex} = 480 nm; λ_{em} = 515 nm) was recorded every 2 min for 90 min using a BioTek Synergy H1 plate reading fluorometer. Data shown are representative traces with values normalized to time zero, $n \geq 6$.

Data Analysis and Presentation

For quantitative analysis of isolated vacuole micrographs, the percentage of boundaries where FM4-64 dye was present but GFP fluorescence was absent was calculated using docked-only vacuoles. Cumulative probability curves were calculated by measuring GFP fluorescence intensities on the boundary and outer membranes of docked vacuoles using an ROI of 4 $\times$ 4 pixels in diameter. Boundary fluorescence relative to outer membrane fluorescence was divided by 2 to account for two apposed membranes in the boundary. Relative luminal GFP fluorescence was calculated using docked vacuoles by measuring the surface area and the length of the boundary, and GFP fluorescence intensities were measured on the outer vacuole membrane, within the lumen and on the boundary membrane. Comparisons were calculated using Student's two-tailed t test with p values of <0.05 indicating significant differences. Quantitative data were acquired using ImageJ software and plotted using Synergy KaleidaGraph 4.0 software. Figures were prepared using Adobe Illustrator CC.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, seven figures, and seven movies and can be found with this article online at <http://dx.doi.org/10.1016/j.devcel.2016.11.024>.

AUTHOR CONTRIBUTIONS

C.L.B. and E.K.M. conceived the project. M.A.K. performed the homotypic vacuole fusion assay shown in Figures 3 and S2. E.K.M. performed all other experiments and prepared all data for publication. E.K.M. and C.L.B. wrote the paper.

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